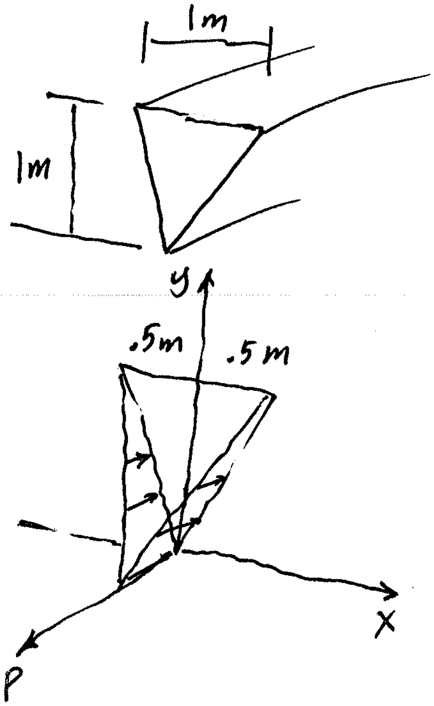
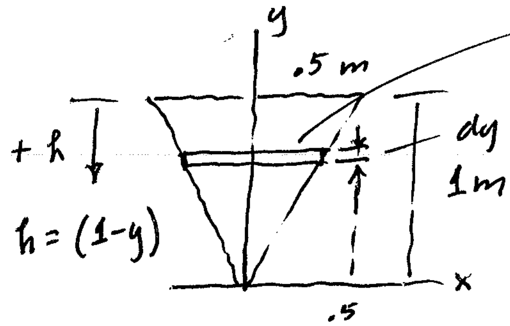
Ex 9.16 Integration



$\rho_w = 1000 \ kg/m^3$



$[dA = 2x \cdot dy]$

$$\begin{bmatrix} P = \gamma \cdot h \\ = \gamma(1-y) \end{bmatrix}$$

$[y = 2x] \rightsquigarrow [x = \frac{y}{2}]$

$dF = dV = p \cdot dA$

Volume of Pressure Distrib

$= \gamma h \, dA$

$= \gamma(1-y)2\left(\frac{y}{2}\right)dy$

$dF = \gamma(y-y^2)\,dy$

$$F = \gamma\int_0^1 (y-y^2)\,dy = 9810\left[\tfrac{1}{2}y^2 - \tfrac{1}{3}y^3\right]_0^1$$

$$= 9810\left[\tfrac{3}{6} - \tfrac{2}{6}\right]$$

$$= \frac{9810}{6}$$

$$\boxed{F = 1.635 \ kN}$$

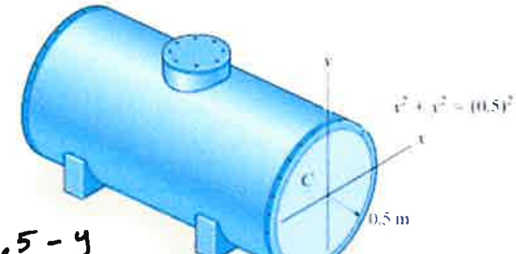
To find $\bar{y}$: Numerator $= \int_V \tilde{y}\,dV = \gamma\int_0^1 y(y-y^2)\,dy$

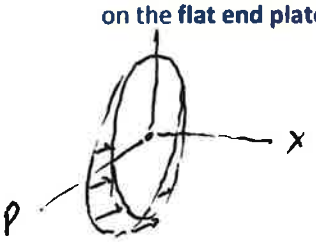
$\tilde{y} = y$ $\quad = 9810\left[\tfrac{1}{3}y^3 - \tfrac{1}{4}y^4\right]_0^1 = 9810\left[\tfrac{1}{12}\right]$

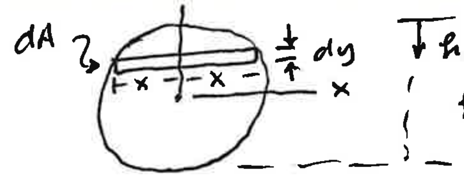
$$\bar{y} = \frac{9810(1/12)}{9810(1/6)} = 6/12 = \boxed{0.5 \ m = \bar{y}}$$

Integral on TI89: $\int((.5-y)*\sqrt{.25-y\text{^}2}), y, -.5, .5) = .19635$

1.5 Determine the **magnitude** of the resultant hydrostatic force acting on the **flat end plate "C"** of the tank. Density of water if 1.0 Mg/m³.







$h = .5 - y$

$P = \gamma h = 9810(.5 - y)$

$x^2 = .25 - y^2$

$x = \sqrt{.25 - y^2}$

$dA = 2x \cdot dy$

$= 2\sqrt{.25 - y^2}\, dy$

$dp = p \cdot dA = 9810(.5-y)(2)\sqrt{}\, dy$

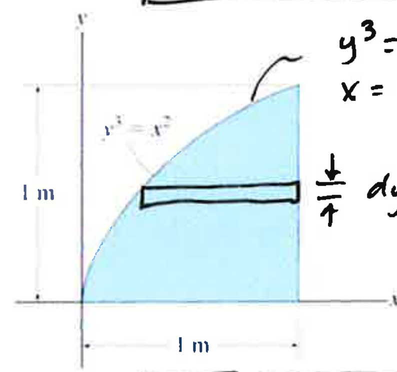
$F_R \int_{-.5}^{.5} 9810(2)(.5-y)\sqrt{.25-y^2}\, dy$

$= 9810(2)(.19635)$

$\boxed{F_R = 3.85 \text{ kN}}$

Section 2 – Moment of Inertia

2.1 Determine the moment of inertia for the shaded about the x axis.



$y^3 = x^2$

$x = y^{3/2}$

$I_x = \int_A \tilde{y}^2 dA$

$dA = (1 - x)\, dy$

$= (1 - y^{3/2})\, dy$

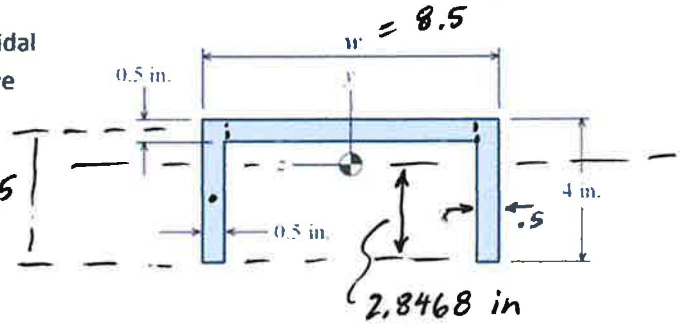
$= \int_0^1 y^2(1 - y^{3/2})\, dy$

$= \int_0^1 (y^2 - y^{7/2})\, dy = \left(\frac{1}{3}y^3 - \frac{2}{9}y^{9/2}\right)_0^1 = \frac{1}{3} - \frac{2}{9} = \boxed{\frac{1}{9}} = \boxed{0.111 \text{ m}^4 = I_x}$

2.2 TRUE FALSE For a general area, the parallel axis theorem would be used when calculating Iy and using a horizontal differential elment. ~ ⊥ to y axis.

$3.75 - 2.847 = .903''$

2.3 If *w* = 8.50 in., find the moment of inertia about the centroidal *z*-axis. The centroid of the shape is located 2.8468 inches above the bottom of the shape.



$I_z = \Sigma(\bar{I} + Ad^2) =$

$= \left[\frac{1}{12}(7.5)(.5)^3 + 7.5(.5)(.903)^2\right]$

$+ 2\left[\frac{1}{12}(.5)(4)^3 + 4(.5)(.847)^2\right]$

$I_z = 3.136 + 2(4.1015)$

$\boxed{I_z = 11.34 \text{ in}^4}$

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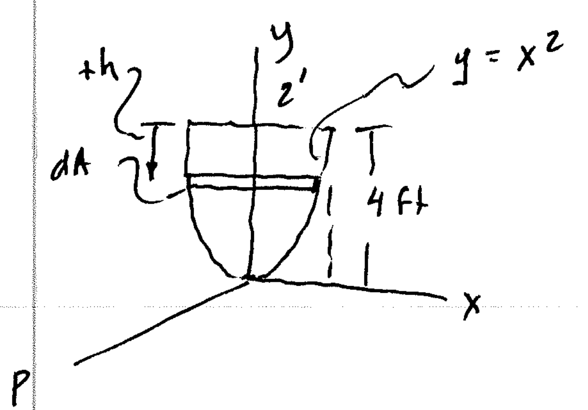

$$dA = (2x)\,dy$$

$$y = x^2 \Rightarrow x = y^{1/2}$$

$$dA = 2y^{1/2}\,dy$$

$$p = \gamma \cdot h$$

$$h = (4-y)$$

$$p = \gamma \cdot (4-y)$$

$$dF = dV = p \cdot dA$$

$$= \gamma(4-y) \cdot 2y^{1/2}\,dy$$

$$F = \int_0^4 \gamma\left(8y^{1/2} - 2y^{3/2}\right)dy$$

$$= 25\left[\tfrac{2}{3}8y^{3/2} - \tfrac{2}{5}2y^{5/2}\right]_0^4$$

$$= 25\left[42.67 - 25.6\right]$$

$$\boxed{F = 426.7 \text{ lb}}$$

Problem: what is γ?

$p = 100 = 4 \cdot \gamma$

$\gamma = 25$ lb/ft^3

compare to water $\gamma = 62.4$ lb/ft^3!

Location, $\bar{y}$? $\tilde{y} = y$

y!

Numerator: $= \int_V \tilde{y}\,dV = \gamma \int_0^4 y\left(8y^{1/2} - 2y^{3/2}\right)dy$

$$= \gamma \int_0^4 \left[8y^{3/2} - 2y^{5/2}\right]dy$$

$$= 25\left[\tfrac{2}{5}8y^{5/2} - \tfrac{2}{7}2y^{7/2}\right]_0^4$$

$$= 25\left[102.4 - 73.14\right] = 731.5$$

$$\bar{y} = \frac{731.5}{426.7} = \boxed{1.714 \text{ ft} = \bar{y}}$$